

1. The signal from a pulsed, polarized source is measured to have an arrival time delay that varies with frequency as $dt_p/d\omega = 1.1 \times 10^{-5} \text{ s}^2$, and a Faraday rotation that varies with frequency as $d\Delta\psi/d\omega = 1.9 \times 10^{-4} \text{ s}$. The measurements are made around the frequency $\omega = 10^8 \text{ s}^{-1}$, and the source is at unknown distance from the earth. Find the mean magnetic field in the interstellar space between the earth and the source.

2. In the note of lecture 2, on page 17, we show that the index of refraction (n) and the absorption coefficient (κ) in dissipative media can be given in the the following:

$$\begin{aligned} n\kappa &= \sqrt{\epsilon\mu} \sqrt{\frac{1}{2} \left(\sqrt{1 + \left(\frac{4\pi\sigma}{\epsilon\omega} \right)^2} - 1 \right)} \\ n &= \sqrt{\epsilon\mu} \sqrt{\frac{1}{2} \left(\sqrt{1 + \left(\frac{4\pi\sigma}{\epsilon\omega} \right)^2} + 1 \right)} \end{aligned}$$

Please derive them.

3. What is the typical plasma frequency for the ionosphere in the atmosphere around the earth? Find the corresponding electron density that gives rise to that plasma frequency. How does the plasma frequency change from day to night?

4. A 100 % linearly polarized interstellar source is 3 kpc away. The average electron density in the direction of this source is 0.03 cm^{-3} . The magnetic field along the line of sight direction, $B_{||}$, is $3\mu\text{G}$. What is the change in the angle of polarization at 100 MHz, and at 1GHz. Express this change on the Poincare sphere assuming that the wave initially has a value of $V = 0$, propagating in the z-axis with a plane of polarization in the x-axis.

5. The Stokes Parameters are defined as (see lecture note 3 page 13)

$$\begin{aligned} S_0 &= I = A_\xi^2 + A_\eta^2 \\ S_1 &= Q = S_0 \cos 2\chi \cos 2\psi \\ S_2 &= U = S_0 \cos 2\chi \sin 2\psi \\ S_3 &= V = S_0 \sin 2\chi \end{aligned}$$

Prove that these parameters can be rewritten with the observable quantities (see page 14) as

$$\begin{aligned} S_0 &= I = A_x^2 + A_y^2 \\ S_1 &= Q = A_x^2 - A_y^2 \\ S_2 &= U = 2A_x A_y \cos \delta \\ S_3 &= V = 2A_x A_y \sin \delta \end{aligned}$$

In reality, wave is not monochromatic but has a finite frequency width and is thus partially polarized. Calculate the degree of polarization if the wave (light) from a source is found to have $I = 80$, $Q = 10$, $U = 15$ and $V = 20 \text{ mJy}$.